

# Electromagnetism 2

## (spring semester 2026)

### Lecture 9

### Diamagnetism and paramagnetism

- ❖ Magnetic properties of atoms
- ❖ Magnetisation and magnetic susceptibility
- ❖ Microscopic theory of diamagnetism and paramagnetism

# Previous lecture

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- ❖ The electric displacement field is defined as

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

- ❖ Gauss law for the electric displacement field:

$$\nabla \cdot \vec{D} = \rho_f \quad \text{or equivalently} \quad \int_S \vec{D} d\vec{S} = \int_V \rho_f dV$$

- ❖ For linear, isotropic and homogeneous (e.g. most) dielectrics,

$$\vec{D} = (1 + \chi_E) \cdot \epsilon_0 \vec{E} = \epsilon \epsilon_0 \vec{E} \quad (\epsilon > 1)$$

- ❖ Boundary conditions (no free surface charges):

$$D_{1n} = D_{2n} \quad \text{and} \quad E_{1t} = E_{2t}, \quad \text{leading to} \quad \frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_1}{\epsilon_2}$$

- ❖ Energy density of the electric field:  $w = \frac{1}{2} \vec{E} \cdot \vec{D}$

# Magnetic properties of atoms (1)

Magnetism is a fundamentally quantum effect: magnetisation phenomena are absent in classical physics.

## Bohr's semiclassical model

Angular momentum of the electron

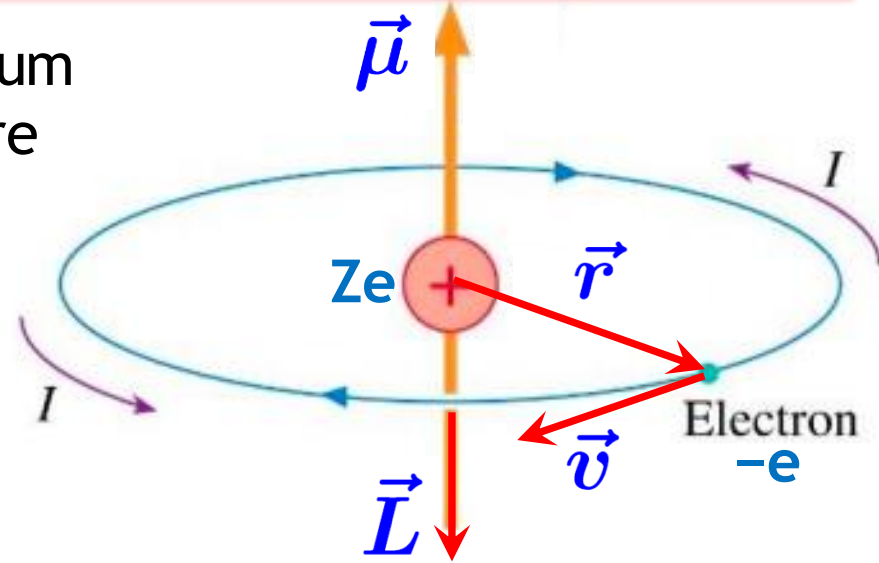
$$\vec{L} = m\vec{r} \times \vec{v}$$

Magnetic dipole moment of a current loop (*lecture 5*)

$$\mu = IS = -\frac{e}{T} \cdot \pi r^2 = -e \frac{v}{2\pi r} \cdot \pi r^2 = -\frac{evr}{2}$$

(**T**: rotation period; **S**: area of loop; **e>0**: elementary charge)

**Gyromagnetic ratio** of electron orbital motion:  $\Gamma = \frac{\mu}{L} = -\frac{e}{2m_2}$



# Magnetic properties of atoms (2)

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Quantisation of angular momentum component:  $L = n\hbar$ ,  $n \in \mathbb{N}$

Quantisation of the orbital magnetic moment of electron:

$$\mu = \frac{e\hbar}{2m_e} \cdot n$$

Bohr magneton:  $\mu_B = \frac{e\hbar}{2m_e} = 9.27 \times 10^{-24} \text{ A} \cdot \text{m}^2$

Electron spin (intrinsic angular momentum):  $S = \pm\hbar/2$

The associated magnetic moment is  $\mu_{\text{spin}} \approx \mu_B$ , therefore the gyromagnetic ratio is  $\sim 2$  times larger than for orbital motion:

$$\Gamma_{\text{spin}} \approx -e/m_e.$$

# Types of magnetism

Magnetism is linked to the charged constituents in materials:

- 1) orbital angular momenta of electrons;
- 2) spin angular momenta of electrons;
- 3) spin angular momenta of nuclei (a much smaller contribution).

In external magnetic field, materials acquire non-zero *magnetisation*  $\vec{M}$ , i.e. induced magnetic dipole moment per unit volume,

$$\vec{M} = \frac{1}{\Delta V} \sum_{\Delta V} \vec{\mu}_i \quad (\text{sum over molecules; unit: [A/m]})$$

For linear, isotropic and homogeneous materials, the *magnetic susceptibility*,  $\chi_B$ , is defined as  $\vec{M} = \chi_B \frac{\vec{B}}{\mu_0}$

Three main types of magnetic materials.

- ❖ *Diamagnetic*:  $\vec{M}$  is opposite to external  $\vec{B}$ ;  $\chi_B < 0$  (and  $|\chi_B| \ll 1$ ).
- ❖ *Paramagnetic*:  $\vec{M}$  is parallel to external  $\vec{B}$ ;  $\chi_B > 0$  (and  $|\chi_B| \ll 1$ ).
- ❖ *Ferromagnetic*: very large, non-linear magnetisation  $\vec{M}$ .

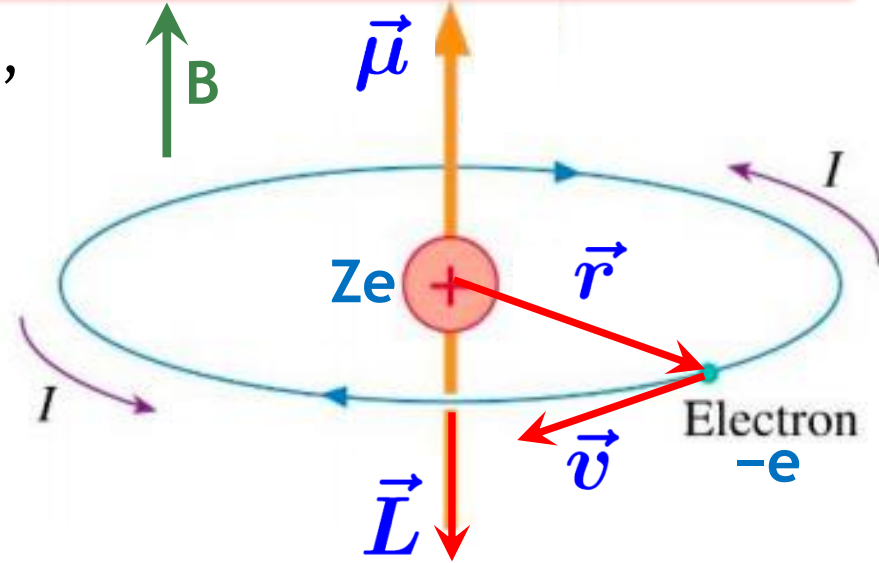
Permanent magnets are magnetised in the absence of external  $\vec{B}$  field. 4

# Diamagnetism (1)

In a varying magnetic field ( $\vec{B} \parallel \vec{\mu}$ ), atomic electrons experience forces due to the induced electric field.

Faraday's law (*lecture 5*):

$$\oint_L \vec{E} d\vec{l} = -\frac{d}{dt} \int_S \vec{B} d\vec{S}$$



**E**-field induced in the electron orbit along the direction of motion:

$$2\pi r E = \pi r^2 \frac{dB}{dt}, \text{ therefore } E = \frac{r}{2} \cdot \frac{dB}{dt} \quad (\text{up to the sign})$$

Equation of motion of the electron:

$$\underbrace{\tau = eEr}_{\text{Torque}} = \underbrace{m_e r^2}_{\text{Moment of inertia}} \frac{d\omega}{dt}, \text{ therefore } \frac{er^2}{2} \frac{dB}{dt} = m_e r^2 \frac{d\omega}{dt}$$

# Diamagnetism (2)

If the magnetic field varies from **0** to **B** over a time **t**, the change of angular frequency of electron rotation is

$$\Delta\omega = \int_0^t \frac{d\omega}{dt} \cdot dt = \frac{e}{2m_e} \int_0^t \frac{dB}{dt} \cdot dt = \frac{eB}{2m_e}$$

This quantity is also known as the *Larmor angular frequency*:

$$\omega_L = \frac{eB}{2m_e} = -\Gamma B$$

Lenz rule: the induced magnetic moment  $\vec{\mu}$  is antiparallel to  $\vec{B}$ .

Therefore, the induced angular momentum  $\vec{L}$  is parallel to  $\vec{B}$ .

$$\vec{\mu} = \Gamma \vec{L} = -\frac{e}{2m_e} \cdot m_e \Delta\vec{\omega} r^2 = -\frac{e}{2m_e} \cdot m_e \frac{e\vec{B}}{2m_e} r^2 = -\frac{e^2 r^2}{4m_e} \vec{B}$$

*Diamagnetism* ( $\chi_B < 0$ ) is a universal phenomenon, and is a consequence of electromagnetic induction.

Diamagnetic materials are (weakly) repelled by magnetic fields. 6

# Diamagnetism, via Lorentz force

[Not discussed in the lecture]

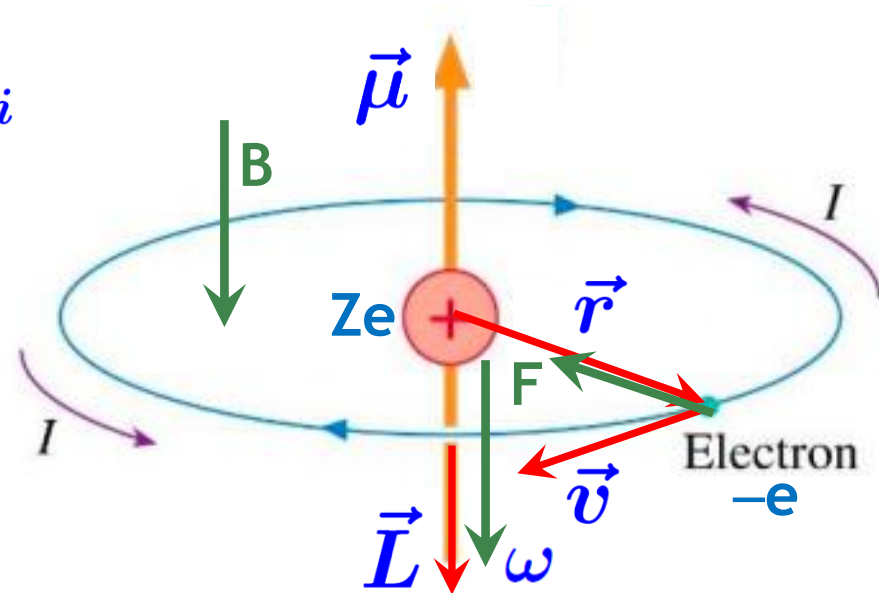
For each electron,  $\vec{\mu}_i = -\frac{e}{2m_e}\vec{L}_i$

## Diamagnetic materials:

electrons of atoms/molecules have zero total angular momentum, i.e. no permanent magnetic dipole moment

In the absence of external field, Newton's second law:

$$m_e a = m_e \omega_0^2 r = \frac{Ze^2}{4\pi\epsilon_0 r^2} ; \quad \omega_0 = \left( \frac{Ze^2}{4\pi\epsilon_0 m_e r^3} \right)^{1/2}$$





# Diamagnetism, via Lorentz force

[Not discussed in the lecture]

In external magnetic field, Lorentz force  $\vec{F} = -e\vec{v} \times \vec{B}$

Newton's second law:

$$m_e \omega^2 r = \frac{Ze^2}{4\pi\epsilon_0 r^2} + e\omega r B; \quad \omega^2 - \frac{eB}{m_e} \omega - \omega_0^2 = 0$$

Then  $\omega^2 - 2\omega_L \omega - \omega_0^2 = 0$

The positive solution of this quadratic equation:

$$\omega = \omega_L + \sqrt{\omega_0^2 + \omega_L^2} \approx \omega_L + \omega_0 \left( 1 + \frac{1}{2}(\omega_L/\omega_0)^2 \right) = \\ \omega_0 + \omega_L \left( 1 + \frac{\omega_L}{2\omega_0} \right) \approx \omega_0 + \omega_L$$

If  $\vec{\omega}$  is in the direction of  $\vec{B}$  it *increases* by  $\omega_L$ .

If  $\vec{\omega}$  is in the opposite direction, it *decreases* by  $\omega_L$ .

# Magnetisation of diamagnetics

Induced magnetic moment for a single electron:  $\vec{\mu} = -\frac{e^2 r^2}{4m_e} \vec{B}$

For an atom,  $\vec{\mu}_{\text{atom}} = -\frac{Ze^2 r_0^2}{6m_e} \vec{B}$

(**Z**: atomic number; **r**<sub>0</sub>: root-mean-square radius of electron orbit;  
**1/6** comes from  $\overline{x^2} + \overline{y^2} = \frac{2}{3} \cdot \overline{R^2}$ , due to the spherical symmetry).

**Magnetisation**: induced magnetic dipole moment per unit volume

$$\vec{M} = -\frac{nZe^2 r_0^2}{6m_e} \vec{B} \quad (\text{n: density of atoms [m}^{-3}\text{)})$$

Magnetic susceptibility for diamagnetics:  $\chi_B = -\frac{\mu_0 nZe^2 r_0^2}{6m_e}$

- ❖ Depends on density only (not on temperature or pressure).
- ❖ Measurements of  $\chi_B$  are used to determine atomic sizes.  
E.g. neon at STP:  $\chi_B = -3.76 \times 10^{-9}$  leads to  $r_0 = 0.49 \times 10^{-9} \text{ m}$ .

# Paramagnetism in monatomic gases

## Paramagnetic materials:

non-zero atomic permanent magnetic dipole moments  
(e.g. odd number of electrons; ions and ionic molecules).

Permanent magnetic moments tend to align with the field (*lecture 5*)

Typical atomic magnetic dipole moments are of the order  $\mu_B$

For a very strong field ( $B=10\text{ T}$ ), work required  
to revert the direction of magnetic moment  $\mu$ :

$$W \approx 2\mu_B B = 2 \cdot 9.27 \times 10^{-24} \text{ A} \cdot \text{m}^2 \cdot 10 \text{ T} \sim 10^{-3} \text{ eV}$$

Much smaller than thermal motion energy at room temperature:

$$\frac{3}{2}kT \approx 0.04 \text{ eV}$$

Polarisation of polar dielectrics (*lecture 7*):  $\vec{P} = \frac{np^2}{3kT} \vec{E}$

Exact analogy for paramagnetic materials:  $\vec{M} = \frac{n\mu^2}{3kT} \vec{B}$

# Magnetisation of paramagnetics

Magnetic susceptibility for *paramagnetic materials*:

$$\chi_B = \mu_0 n \left( \underbrace{\frac{\mu^2}{3kT}}_{\substack{\text{Paramagnetic term} \\ \text{due to alignment of permanent} \\ \text{atomic magnetic moments } (\mu)}} - \underbrace{\frac{Ze^2 r_0^2}{6m_e}}_{\substack{\text{Diamagnetic term} \\ \text{due to induced} \\ \text{magnetic moments}}} \right) > 0$$

- ❖ Materials with  $\mu=0$  are usually diamagnetic.
- ❖ For most materials with  $\mu \neq 0$ , the paramagnetic term dominates over the diamagnetic term.
- ❖ This leads to the *Curie's law* ( $\chi_B \sim 1/T$ ) for gaseous paramagnetics at  $\mu B \ll kT$  (i.e. sufficiently large  $T$ ).
- ❖ Paramagnetic materials are (weakly) attracted by magnetic fields.

# Summary

- ❖ External magnetic fields induce magnetisation of materials, which is characterised by the *magnetic susceptibility*  $\chi_B$ :

$$\vec{M} = \chi_B \frac{\vec{B}}{\mu_0}$$

- ❖ *Diamagnetism* ( $\chi_B < 0$ ) is a universal phenomenon: *induced magnetic dipole moments* are directed against the **B** field.
- ❖ *Paramagnetism* ( $\chi_B > 0$ ) is due to alignment of *permanent atomic magnetic moments* along the **B** field.
- ❖ Microscopic theory of magnetism leads to

$$\chi_B = \mu_0 n \left( \frac{\mu^2}{3kT} - \frac{Ze^2 r_0^2}{6m_e} \right)$$